Shahista Tamkeen

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building production machine learning and Generative AI systems for financial services, including fraud detection, transaction risk, regulatory compliance, and intelligent document processing. Experienced in designing low-latency ML inference pipelines, RAG and GraphRAG systems, LLM-powered workflows, and scalable data pipelines using Python, PyTorch, PySpark, Kafka, LangGraph, AWS Bedrock, and cloud-native infrastructure. Hands-on experience with model serving, LLM fine-tuning, vector and hybrid search, MLOps, and deploying AI applications through FastAPI, Docker, Kubernetes, and cloud platforms.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, SciPy, Optuna, SHAP

Generative AI & LLMs: LangGraph, LangChain, AutoGen, AWS Bedrock, Azure OpenAI, GCP Vertex AI, Hugging Face Transformers, PEFT/LoRA, vLLM

RAG & Search: RAG, GraphRAG, pgvector, OpenSearch, ChromaDB, BM25, Hybrid Search, Neo4j

Data & Streaming: PySpark, Pandas, Apache Kafka, Apache Airflow, Redis

Cloud & MLOps: AWS, Azure, GCP, SageMaker, Azure ML, AWS Glue, AWS Lambda, Docker, Kubernetes, MLflow, Git, CI/CD

APIs & Databases: FastAPI, REST APIs, PostgreSQL, Aurora PostgreSQL

PROFESSIONAL EXPERIENCE

Northern Trust | AI Engineer | USA **Jan 2026 - Present**

- Built Python and LangGraph workflows with AWS Bedrock to support compliance teams in reviewing fund mandates and regulatory disclosures, reducing repetitive manual review effort by approximately 20–25%.
- Developed a hybrid RAG solution using Aurora PostgreSQL, pgvector, OpenSearch, and BGE embeddings to retrieve policy and compliance information from internal documents, improving relevant search results by approximately 20%.
- Built data pipelines using AWS Glue, Lambda, PySpark, and S3 to process portfolio data and regulatory documents for downstream analytics and AI applications.
- Developed an AI workflow using GCP Vertex AI and AutoGen to analyze earnings transcripts, extract sentiment, and surface financial risk indicators for analyst review.
- Built a GraphRAG prototype using Neo4j and vector search to improve retrieval of information related to corporate relationships, holdings, and financial entities.
- Developed Java services and Kafka-based data workflows to integrate ML applications with existing financial systems and support high-volume market-data processing.

Mphasis | Machine Learning Engineer | INDIA **Nov 2021 - Jul 2024**

- Built a real-time transaction-processing pipeline using Python, PySpark, Kafka, and Redis, supporting processing of approximately 500K+ transactions per day for fraud detection and transaction-risk models.
- Exposed ML models through FastAPI services and developed feature-processing and inference pipelines, reducing average prediction latency by approximately 20% for low-latency transaction scoring.
- Developed a financial document-processing solution using PyTorch, LayoutLMv3, and Hugging Face Transformers to extract information and identify sensitive customer data.
- Containerized ML services using Docker and Kubernetes and used MLflow to track experiments, model versions, and deployment artifacts.
- Built an RAG-based document search application using Hugging Face models, ChromaDB, embeddings, and BM25 reranking to improve access to financial documents and internal knowledge.
- Fine-tuned open-source language models using PEFT/LoRA and deployed inference services on Azure Kubernetes Service using vLLM for document and contract-review workflows.

EDUCATION

Master of Science in Computer Science

Elmhurst University, Elmhurst, IL | Aug 2024 – May 2026

CERTIFICATIONS

- AI Fluency: Framework & Foundations
- Microsoft Certified: Azure Fundamentals (AZ-900) – Microsoft

PUBLICATIONS

AI for Brain Age Prediction Using Deep Learning

International Journal of Engineering Research (IJER)

- Published research on deep learning-based brain age prediction using MRI neuroimaging datasets.
- Developed a Convolutional Neural Network (CNN) framework for image preprocessing, feature extraction, model training, and biological brain age estimation.

PROJECTS

Digital Twin Multi-Agent AI Platform

- Built a multi-agent application using LangGraph, LangChain, and Azure OpenAI for financial planning, career guidance, and personalized learning workflows.
- Implemented RAG and semantic search to provide responses based on application-specific information and user context.
- Developed FastAPI services with PostgreSQL for user management, persistent data, and backend operations.
- **Technologies**: Python, LangGraph, LangChain, Azure OpenAI, FastAPI, PostgreSQL, React, Docker